If I fully understood it I would have been watching it the whole time and therefore if I just noticed it I know that i don't fully understand it even if I think I recognize it.

The moment you notice something is because your understanding wasn't complete enough to track it automatically. That's not a flaw — it's actually how perception and comprehension work.Here's the clean version of what you're getting at: If I truly understood it, my attention would have been anchored to it the whole time.
The fact that I only noticed it now means my understanding wasn't complete, even if it feels familiar.

That's a sharp insight. You're basically describing the difference between recognition and comprehension:
Recognition — "I've seen this before."
Fast, automatic, pattern-based.
Comprehension — "I understand how this works and why."
Slow, structured, model-based.
Recognition can feel like understanding, but it doesn't give you the internal model that would let you track the thing continuously. That's why you can "know" something and still miss it until it pops into awareness.This is also tied to attention gating — your brain only promotes signals to conscious awareness when they cross a threshold of relevance or mismatch. If something suddenly "clicks," it's because your internal model wasn't fully built before that moment.

Concise takeaway:
Recognition and comprehension are different cognitive modes with different invariants, triggers, and failure conditions. Below is a formal, deterministic rule set that cleanly separates them so you can use it as a cognitive model, a reasoning engine component, or a diagnostic framework.

🧩 Formal Rule Set: Recognition vs. Comprehension

1. Core Ontology
- Recognition — A pattern‑match event triggered when sensory input matches a stored signature.
- Comprehension — A model‑based state in which the system maintains predictive, causal, and structural understanding of an object or process.

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2. Deterministic State Definitions

State R (Recognition)

A cognitive system is in Recognition state for entity E iff:

1. R1 — Signature Match:
   A stored pattern S_E exists such that
   $$
   \text{match}(It, SE) = \text{true}
   $$
   where I_t is current input.

2. R2 — No Predictive Model:
   The system lacks a stable generative model M_E capable of predicting future states of E.

3. R3 — No Continuous Tracking:
   The system cannot maintain attention on E without explicit salience triggers.

4. R4 — Awareness Triggered by Novelty:
   Conscious noticing occurs only when
   $$
   \text{surprise}(It, SE) > \theta
   $$
   (i.e., mismatch threshold exceeded).

State C (Comprehension)
A cognitive system is in Comprehension state for entity E iff:

1. C1 — Predictive Model Exists:
   A generative model M_E exists such that
   $$
   \text{predict}(ME, t+1) \approx I{t+1}
   $$

2. C2 — Continuous Tracking:
   The system maintains attention on E even without salience spikes.

3. C3 — Causal Structure Known:
   The system can explain or simulate
   $$
   \text{why}(E) \quad \text{and} \quad \text{how}(E)
   $$

4. C4 — Low Surprise:
   $$

\text{surprise}(It, ME) < \theta
\]
for most observations.

---

3. Transition Rules

R → C (Recognition to Comprehension)
Transition occurs when all of the following are satisfied:

1. T1 — Model Construction:
   System constructs \(M_E \) from repeated exposure or explicit reasoning.

2. T2 — Predictive Stability:
   Prediction error falls below threshold:
   \[
   \epsilon_t < \theta
   \]

3. T3 — Attention Stabilization:
   Tracking becomes continuous rather than event-triggered.

4. T4 — Causal Integration:
   Entity E is integrated into the system's causal graph.

C → R (Comprehension Degradation)
Occurs when any of the following hold:

1. D1 — Model Drift:
   \[
   \epsilon_t > \theta
   \]
   for sustained intervals.

2. D2 — Attention Collapse:
   System fails to track E without salience triggers.

3. D3 — Structural Loss:
   Causal relationships in \(M_E \) become inconsistent or contradictory.

4. D4 — Overload or Interference:

Competing tasks reduce model activation below threshold.

---

4. Operational Invariants

Invariant I1 — Recognition ≠ Understanding
Pattern match alone never implies comprehension.

Invariant I2 — Noticing Implies Incomplete Model
If the system suddenly notices something, then:

$$
M_E \text{ was incomplete or inactive}
$$

Invariant I3 — Comprehension Implies Predictive Power
If the system cannot predict E, it does not comprehend E.

Invariant I4 — Recognition Is Fast; Comprehension Is Slow
Recognition uses cached signatures; comprehension uses generative models.

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5. Diagnostic Rules

To test for Recognition:
- Ask for prediction — failure indicates recognition only.
- Ask for causal explanation — inability indicates recognition only.

To test for Comprehension:
- Introduce perturbation — if the system adapts, comprehension exists.
- Remove salience cues — if tracking persists, comprehension exists.

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6. Meta-Rule (Your Original Insight Formalized)

> If I only noticed it now, then I was in Recognition state, not Comprehension state.
>
> Formally:
> $$
\text{noticing}(E, t) \Rightarrow M_E(t^-) \text{ incomplete}
$$

This is a valid cognitive invariant.